

Options Monitor

Powered by iVolatility

The Options Monitor, powered by iVolatility, allows users to explore quotes for stocks, ETFs and indexes, while viewing a variety of key options data, including the Greeks, implied volatility, bid-ask spreads and volume. Expirations can be changed to display additional data, including LEAPS®, weekly options and quarterlies. Plus, access to the Options Calculator and Probability Calculator is available directly from within the monitor. Note that all data for US stocks, options, dividends and interest rates are provided with a 20-minute delay. If the markets are closed, the previous trading day's market closing price will be used.

Choose a Stock Symbol, Index Symbol, or Option Symbol

Input any stock, index or option symbol. If you don't know the symbol or the option you can browse by company name, instrument, market or exchange via the "Symbol Lookup" icon. Once you select a symbol, data will populate the rectangular box to the right of the search box, as well as the horizontal list of all available option expirations below.

[SYMBOL]

Search instrument

Expirations

All

Nov03 '23(W)

26.59

10.55

Nov06 '23(W)

11.60

0.55

Nov07 '23(W)

12.23

0.50

Nov08 '23(W)

12.56

0.39

Price

\$ 434.77

Open

433.14

Hi

434.91

Lo

433.01

Expirations

All

Mid price

434.78

5

Data Displayed

To the right of the search box, various metrics for the selected symbol are displayed. Hover over each for a definition and additional detail on how those metrics are gathered and calculated. Available Metrics include:

- Price / Open / High / Low
- Implied Volatility Index (IVX) 30 / IVX 30 Chng / Implied Volatility Percentile (IVP) 30 /
- Historical Volatility (HV) 20
- Options Volume (OV) / AvgOV1MO / Call OV / Put OV
- Open Interest (OI) / Avg OI 1Mo / Shares Outstanding (SO) / Share Volume (SV)

Available expirations will be listed horizontally below this large box and each expiration date is clickable to zoom in on that month individually, or the user can click the 'Show All' expiration dates to have all months appear in the table below.

See glossary in appendix for more.

[SYMBOL]

Search instrument

Price\$ 434.77

Open433.14

Hi434.91

Lo433.01

IVX 30 (%)13.41

IVX30Chg%

-0.45(-3.25%)

IVP 30 (%)28.4

HV 20 (%)15.8

OV1.46M

AvgOV1MO9.31M

C OV670,603

P OV793,371

OI20,499,695

AvgOI1MO19.75M

SO931.58M

SV11.74M

Expirations

Show all

Nov03 '23(W)

Nov06 '23(W)

Nov07 '23(W)

Nov08 '23(W)

Nov09 '23(W)

Nov10 '23(W)

Nov13 '23(W)

Nov14 '23(W)

Nov15 '23(W)

Nov16 '23(W)

Nov17 '23(W)

26.59

11.60

12.23

12.56

12.92

13.12

12.20

12.86

13.05

13.35

13.47

10.55

0.55

0.50

0.39

0.28

0.20

0.12

0.14

0.07

0.09

0.05

OIC®

occ

Data Displayed Continued...

The drop downs below this display of available expiration months allows for further customization of the display including:

- Expirations
- Mid Price
- Number of strikes represented per month
- Calls & Puts

The green gear icon lists viewable column data for the table below. It includes metrics such as the Greeks, implied volatility levels, previous close, size, time, etc. All of these can be toggled on or off to be added or subtracted from the view in the table below.

The below table will then populate data for each month of options and display the pricing matrix with information as selected by the user in the settings.

The header for each month includes these defaults:

- Expiry date of the option
- DTE – how many days are left before the option expires
- Forward price – the forward price of the underlying being used to calculate the option values for that month.

Expiry dates	Nov03 '23(W)	Nov06 '23(W)	Nov07 '23(W)	Nov08 '23(W)	Nov09 '23(W)	Nov10 '23(W)
Show all	24.17 8.31	11.24 0.35	12.07 0.50	12.59 0.61	13.01 0.53	13.23 0.49
Expirations	All	Mid price	433.67	5	Call/Put	
C Vega	C Delta	C ImpVol (%)	C Volume	C Bid	C Ask	Strike
Expiry: 2023-11-03 Days: 0 Forward: 433.72						

Mid price	433.67	5	Call/Put	
Column Settings				
0	Forward	Apply	Select all	Unselect all
23.43				
23.82		Restore default columns		
22.65	Search...			
22.23	Alpha			
21.72	Ask			
3	Ask Size			
11.64	Ask Time			
11.33	Bid			
10.96	Bid Size			
10.83	Bid Time			
10.59				
4	Forward: 433.69			

Expiry dates	Nov03 '23(W)	Nov06 '23(W)	Nov07 '23(W)	Nov08 '23(W)	Nov09 '23(W)	Nov10 '23(W)	Nov13 '23(W)	Nov14 '23(W)	Nov15 '23(W)	Nov16 '23(W)	Nov17 '23(W)
Show all	24.17 8.31	11.24 0.35	12.07 0.50	12.59 0.61	13.01 0.53	13.23 0.49	12.24 0.32	12.90 0.35	13.08 0.28	13.33 0.24	13.47 0.20
Expirations	All	Mid price	433.67	5	Call/Put						
C Vega	C Delta	C ImpVol (%)	C Volume	C Bid	C Ask	Strike	P Bid	P Ask	P Volume	P ImpVol (%)	P Delta
Expiry: 2023-11-03 Days: 0 Forward: 433.84											
0.04	0.63	24.06	48,291	1.87	1.9	432.5	0.53	0.54	37,200	24.57	-0.31
0.04	0.63	24.06	48,291	1.52	1.53	433	0.68	0.69	144,043	24.02	-0.37
0.04	0.48	23.13	142,839	0.93	0.94	434	1.09	1.1	129,290	23.09	-0.52
0.04	0.32	22.55	176,366	0.51	0.52	435	1.67	1.68	45,272	22.51	-0.68
0.02	0.18	21.85	151,022	0.24	0.25	436	2.27	2.28	10,560	21.75	-0.82

Calculators

At any point, a user may click on the green calculator symbol on each strike to bring up the options and probability calculator function for that option strike. The fields in the calculator will be pre populated with data from the strike used to activate the calculator.

C Ask	Strike	P Bid	P Ask
1.9	432.5	0.52	0.53
1.55	433	0.67	0.68
0.95	434	1.09	1.1
0.52	435	1.64	1.65
0.26	436	2.34	2.35

See Options Calculator and Probability Calculator for more information.

Glossary

ALPHA: Alpha is an option Greek. Alpha is a ratio of Gamma over Theta. Thus Alpha indicates the relative value of owning gamma relative to the current level of theta. It is a measure that allows for comparison of several different options on the basis of how much they cost daily to own (daily Theta) versus the potential Gamma derived return (profits from movement) from owning them. The greater absolute value of Alpha the more potential for profits exists against the loss from Theta for long positions. The converse is true for short positions.

AVERAGE OF PUTS AND CALLS IV: This number is derived internally by finding a simple average or a mean of the put and call implied volatility.

CALL BID/ ASK MEAN: This number is calculated by finding a simple average of the market maker's bid and ask price for call options only.

CALL/ PUT VOLATILITY RATIO: This is a variation of the traditional volume based Put/Call Ratio. However, in this particular case, the Put/Call Ratio is based on a Vega weighting of implied volatility. The ratio is arrived at by dividing the Vega weighted implied call volatility by the Vega weighted implied put volatility. This helps express the bullish/bearish sentiment of market participants.

CHANGE VS. YESTERDAY: When in the implied Volatility section, this number can be derived by comparing the implied volatility index last figure to its previous night's level. However, when in the historical volatility section this number is derived by taking the historical volatility last figure and comparing to its previous night's level. In both cases, the differences are then expressed as simple numerical changes.

% CHANGE VS. YESTERDAY: The IV index % change number is derived by comparing the implied volatility index last figure shown on the screen against the previous night's level, this difference is then expressed as a percent. In the case of HV % change, this number is calculated by taking the difference between the historical volatility last figure shown on the screen and the previous night's level. The difference is then expressed as a percentage.

DELTA: Delta is an option Greek. The theoretical amount an option's price will change for a corresponding one-point change in the price of the underlying security.

GAMMA: Gamma is an option Greek. The theoretical change in Delta divided by the dollar change in the price of the underlying security's price. It is the measurement of the rate of change in an options price with respect to the underlying price.

HISTORICAL VOLATILITY OR HV: Historical volatility is the volatility that actually occurred to the underlying stock over some previous time period. It can be derived in several different fashions. The most common way to calculate historical volatility is to use the daily closing prices and compare them over some past time period.

IMPLIED VOLATILITY OR IV: Implied Volatility is the market's best approximation of future volatility, and it is obtained by plugging the current option price into an option pricing model and backsolving to arrive at the options implied volatility.

IMPLIED VOLATILITY AS A % OF THE LATEST 30 DAY HV LEVEL: This figure is calculated by taking the implied volatility last and dividing it by the 30 day historical level. The resulting figure is then expressed as a percentage.

IMPLIED VOLATILITY INDEX (IV INDEX or "IVX"): IVX is the composite measure of the stock's implied volatility calculated from an option's price using IVolatility proprietary methodology. Example: IVX30 is calculated for the forward 30 calendar days.

OPEN INTEREST: Open interest is defined as the number of outstanding contracts on a particular option class (all put or call contracts on the same underlying stock or index) or series (all options on the same underlying stock or index having the same strike price and expiration date) that are still open. As of the reporting date, these options have not been exercised, closed out or allowed to expire.

OPTION PRICING MODEL: An option model is used to assess an option's price. The Black-Scholes model led the way in the pricing of equity options back in 1973. However, as options trading has grown, various refinements to the Black-Scholes model have been introduced. Nevertheless, most models still incorporate the following factors into their pricing assumptions: the underlying security price, the strike price, the time until expiration, any dividends to be paid, the level of interest rates, and the implied volatility.

OPTION VALUE: This is the option's price as determined by the parameters input into the option model. The inputs consist of the following: the underlying security price, the strike price, the time until expiration, any dividends to be paid, the level of interest rates, and the implied volatility.

PUT BID/ ASK MEAN: This number is calculated by finding a simple average of the market maker's bid and ask spread for put options only.

Glossary Continued...

RHO: Rho is an option Greek. A measure of change in an options price in response to a one percentage point change in the risk-free interest rate.

STRIKE PRICE: The stated price per share for which the underlying stock may be purchased (in the case of a call) or sold (in the case of a put) by the option holder upon exercise of the option contract.

THETA: Theta is an option Greek. A measure of how much an option's price theoretically decays with each passing day with the price of the underlying security and implied volatility remaining unchanged.

VEGA: Vega is an option Greek. The measure of theoretical change in an option's price in response to a one percentage point change in implied volatility.

VOLATILITY: Volatility can be defined as the propensity of the underlying security's market price to change in either direction and is based on the standard deviation of the asset's return. It is quoted as an annualized percentage rate. Volatility is a variable that appears in option pricing formulas and therefore the greater the volatility, the greater the premium of an option. There are several different types of volatility. These include historical, implied, actual, seasonal and forecasted volatility.

VOLUME: Volume is defined as the actual number of stock shares or option contracts that traded on that particular day.

Disclaimer

The Probability Calculator and 20-minute delayed options quotes are provided by IVolatility, and NOT BY OCC. The tool is being provided for educational purposes only. OCC makes no representation as to the timeliness, accuracy or validity of the information and this information should not be construed as a recommendation to purchase or sell a security, or to provide investment advice. For more information or help, please contact us at options@theocc.com.

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